Academic Year: 2025–2026
Date: September 4, 2026

MEMVERSE: Zero-Trust Privacy Gateway for Multimodal AI Memory

A client-governed security architecture reconciling personalization and data privacy in large language models.

1. Project Description

MEMVERSE is a zero-trust privacy gateway designed to enable continuous, personalized AI interactions without exposing raw personal identifiers to external model providers. By enforcing a 12-stage security pipeline with dynamic policy transformations, cryptographic memory passports, and tamper-evident audit ledgers, MEMVERSE ensures that external LLMs receive only the minimum necessary generalized semantic context required to answer user queries across text, structured documents, and facial biometrics.

2. Team Details

Name	Role	Institution & Department
Satvik Kesarwani	Team Lead	Indian Institute of Information Technology (IIIT), Pune B.Tech, Computer Science and Engineering (MIS: `112315166`)
Astha Jain	Core Developer	Institute of Engineering and Technology (IET DAVV), Indore

3. Project Deliverables & Submission Links

1. Project Video Demonstration
Walkthrough video recording showcasing zero-trust memory, multimodal document ingestion, biometric privacy, and live trace inspection:
`https://drive.google.com/file/d/1VjpQrxNEmAkfgZdqcdH9a4bNIQCHAY9H/view?usp=sharing`

2. Live Working Prototype (Frontend Application)
Interactive web client featuring real-time chat, policy exploration, memory passport management, and 12-stage trace inspection:
`https://memverse-satvikkesarwanis-projects.vercel.app`

3. Gateway API Service (Backend)
FastAPI zero-trust gateway service integrated with live NVIDIA NIM models and cryptographic ledger endpoints:
`https://memverse-api.onrender.com/api/status`

4. Source Code Repository
Complete project source code repository including backend services, frontend application, and test suites:
`https://github.com/satvikkesarwani/memverse`

5. Project Details & Demonstration Media Folder
Submission repository containing project demonstration recordings, architecture diagrams, and supporting documentation:
`https://drive.google.com/drive/folders/1UO3jMmCsiOP9CiwQckT6S5NyeJYVo_f9?usp=sharing`